Definition 6.4.5. We call vectors of the form

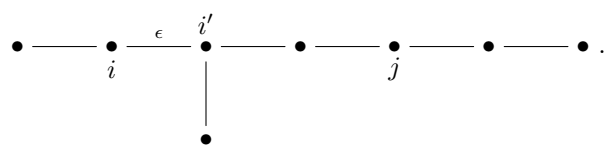
$$\alpha_i = (0, \dots, \overbrace{1}^{i\text{th}}, \dots, 0)$$

simple roots.

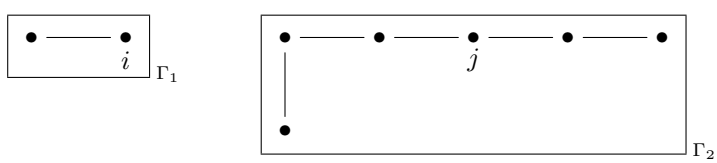
The α_i naturally form a basis of the lattice $\mathbb{Z}^n$.

Lemma 6.4.6. *Let α be a root, $\alpha = \sum_{i=1}^n k_i \alpha_i$. Then either $k_i \geq 0$ for all i or $k_i \leq 0$ for all i.*

Proof. Assume the contrary, i.e., $k_i > 0$, $k_j < 0$. Without loss of generality, we can also assume that $k_s = 0$ for all s between i and j. We can identify the indices i, j with vertices of the graph Γ:



Next, let ϵ be the edge connecting i with the next vertex towards j and let i' be the vertex on the other end of ϵ. We then let Γ_1, Γ_2 be the graphs obtained from Γ by removing ϵ. Since Γ is supposed to be a Dynkin diagram — and therefore has no cycles or loops — both Γ_1 and Γ_2 will be connected graphs which are not connected to each other:



Then we have $i \in \Gamma_1$, $j \in \Gamma_2$. We define

$$\beta = \sum_{m \in \Gamma_1} k_m \alpha_m, \quad \gamma = \sum_{m \in \Gamma_2} k_m \alpha_m.$$

With this choice we get

$$\alpha = \beta + \gamma.$$

Since $k_i > 0, k_j < 0$, we know that $\beta \neq 0, \gamma \neq 0$ and therefore

$$B(\beta, \beta) \geq 2, \quad B(\gamma, \gamma) \geq 2.$$